

# Dot against cross

Two ways of multiplying vectors that disagree on almost every attribute. Where one is zero the other is largest, and between them they measure the two things a pair of vectors has: how much they agree, and how much they span.

| TYPE                                                                             | CONDITION            | OUTPUT                                                                         | DEFINED IN | SWAPPING THE ARGUMENTS | ZERO WHEN                        | GEOMETRIC READING               |                                   |
|----------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------|------------|------------------------|----------------------------------|---------------------------------|-----------------------------------|
| MEASURES AGREEMENT                                                               |                      |                                                                                |            |                        |                                  |                                 |                                   |
|  | <b>Dot product</b>   | $\mathbf{a} \cdot \mathbf{b} = \ \mathbf{a}\  \ \mathbf{b}\  \cos \theta$      | a scalar   | any $\mathbb{R}$       | unchanged — commutative          | perpendicular, $\theta = \pi/2$ | projection of one onto the other  |
| MEASURES SPREAD                                                                  |                      |                                                                                |            |                        |                                  |                                 |                                   |
|  | <b>Cross product</b> | $\ \mathbf{a} \times \mathbf{b}\  = \ \mathbf{a}\  \ \mathbf{b}\  \sin \theta$ | a vector   | only $\mathbb{R}$      | sign reverses — anti-commutative | parallel, $\theta = 0$ or $\pi$ | area of the parallelogram spanned |

The opposition is systematic rather than coincidental. One extracts  $\cos \theta$  and one extracts  $\sin \theta$ , and every difference below follows: zero at opposite angles, scalar against vector, commutative against sign-reversing. Given both products,  $\theta$  is determined completely — neither alone suffices.